

Marcos Esparza

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Education

Bachelor of Science: Mechanical Engineering

University of Texas Permian Basin | GPA: 3.0

Aug 2022 - December 2026

Odessa, TX

Relevant Coursework: Thermodynamics II, Fluid Mechanics II, Mechanics of Materials, Thermo-Fluids Lab, Engineering Design

Activities: FAST, ASME, AIAA, SPE

Technical Projects

Falcon Aeronautics & Space Team (FAST) – IREC 10k COTS Rocket

Aerodynamic Design Lead – Fin Can & Structures | UTPB | Aug 2024 – Present

- Lead aerodynamic and fin-can design for FAST's 10,000-ft COTS rocket entry in the Spaceport America Cup and Lone Star Cup.
- Built SOLIDWORKS models of the airframe and fin can, integrating 98 mm motor, recovery hardware, and avionics bays.
- Performed stability and fin-flutter analysis in OpenRocket and MATLAB, maintaining a static margin of 2.88 calibers and a flutter margin $>1.5\times$ maximum flight velocity.

Thermo-Fluids / Mechanical Systems Lab – UTPB

Undergraduate Lab Engineer | Jan 2024 – Present

- Ran experiments on Pelton turbine, cross-flow heat exchanger, and flow-meter rigs to measure efficiency, head loss, and heat-transfer coefficients.
- Processed pressure/temperature/flow data in MATLAB and Excel to compare experimental results to theory and quantify error sources.
- Wrote ASME-style reports with uncertainty analysis and discussion of system performance, using NIST data and standards for reference.

L2 High-Power Rocket (In-Progress Personal Project)

- Designing and building a 4" dual-deploy cardboard airframe with fiberglass-reinforced sections, 38 mm motor mount, and through-the-wall (TTW) fins for an upcoming NAR/Tripoli L2 certification flight.
- Modeling performance in OpenRocket ($\approx 4,200$ ft apogee on J425) and sizing the recovery system (36" main, 14" drogue) to achieve safe descent rates and meet L2 certification requirements.

Fluid Friction Experiment (H408)

- Wrote ASME-style reports with uncertainty analysis and system-performance discussion, including fluid-friction/bend-loss tests on the TecQuipment H408 (measured pressure drop, head loss, and loss coefficients vs Moody/Blasius correlations).

Experience

Logistics Associate — Academy Sports + Outdoors, Midland, TX

Jul 2024 – Present

- Process and stage online and in-store pickup orders, consistently meeting same-day deadlines.
- Perform cycle counts and resolve stock discrepancies to maintain accurate inventory.

Overnight Deli Production — H-E-B, Midland, TX

Oct 2022 – Jan 2024

- Prepared and stocked deli products overnight to support next-day store operations.
- Helped train new team members on shift procedures and safe handling practices.

Team Member — Whataburger, Midland, TX

Oct 2020 – Mar 2022

- Supported front counter and kitchen operations during high-volume periods while maintaining order accuracy and cleanliness.

Technical Skills

Programming: MATLAB, Python (basic), LabVIEW (exposure)

Software: SOLIDWORKS, OpenRocket, Excel, Word, PowerPoint

Test & Lab: Pressure/temperature instrumentation, flow meters (orifice, venturi, rotameter), Pelton turbine rig, cross-flow HX

Fabrication: Basic machining & hand tools, 3D printing, epoxy layups, high-power rocketry construction (TTW fins, dual-deploy, avionics integration)

Engineering: Aerodynamic & structural design, test data analysis, systems integration, mechanical assembly, technical writing

Certifications: Tripoli/NAR High-Power Rocketry Level 1; Level 2 in progress